Automatic formulation of MILP models from natural language descriptions with expert knowledge and performance-aware configuration

Qingyang Li[1,2][0009-0007-3861-9668], Lele Zhang[1,2][0000-0002-0527-7426], and Vicky Mak-Hau[3][0000-0002-9306-5780]

[1] The University of Melbourne, Melbourne, Australia
[2] ARC Training Centre in Optimisation Technologies, Integrated Methodologies, and Applications (OPTIMA), Melbourne, Australia
qingyang.li5@student.unimelb.edu.au, lele.zhang@unimelb.edu.au
[3] Deakin University, Melbourne, Australia
vicky.mak@deakin.edu.au

1 Introduction and background

Mixed-integer linear programming (MILP) is widely used in decision-support applications in logistics, production, and workforce planning. In many organisations, the main bottleneck is not the solver. It is the modelling step, where requirements written in natural language must be translated into a correct MILP model. This translation is time-consuming, expertise-driven, and error-prone, which limits adoption by teams without dedicated optimisation modellers.

Large language models (LLMs) have enabled rapid progress in generating optimisation models from text, including direct prompting and multi-step systems [4]. However, recent studies still report recurring failure modes in automatic modelling tasks: weak handling of implicit constraints and logical relations, dependence on training data, limited benchmark complexity, and hallucinated or inconsistent constraints [1–3]. These gaps motivate my thesis. The goal is to build reliable systems that produce controllable MILP models from natural language, reduce the time and expertise required for optimisation modelling, and automatically configure advanced solution strategies when large instances remain computationally hard.

My PhD investigates three connected questions, organised as three topics:

1. **Reliable MILP synthesis from unstructured natural language descriptions.** Can we convert free-form natural language problem descriptions into correct MILP formulations, including binary decisions and logical relations, with stable performance across trials?
2. **Expert-knowledge-guided modelling for complex problem types.** Can we apply automatic modelling in richer domains, where key modelling components are implicit and must be derived, while keeping control and traceability of the final formulation?
3. **Performance-aware decomposition configuration.** Can we automatically generate a decomposition configuration (e.g., Benders master problem

and subproblems) that is guided by computational performance rather than by structure alone?

2 Thesis approach and current results

Topic 1: Multistage MILP synthesis from natural language. Topic 1 tackles a practical modelling gap: real specifications often include binary decisions and logic (e.g., “if-then”, “either-or”), which are exactly where end-to-end LLM prompting tends to produce plausible-but-wrong formulations. To improve reliability, this work introduces a constraint-type-driven multistage framework that first recognises the decision variables, then classifies each paragraph as either the objective or a constraint type with a fine-tuned LLM, and finally constructs expressions through type-specific templates plus automatic implicit linking constraints for logical relations.

Beyond the method, Topic 1 contributes a 30-instance MILP problem benchmark that covers binary variables and multiple logic-constraint classes. Across repeated trials, the multistage framework produces near-complete instance-level correctness and remains stable. It outperforms strong automated baselines, with error analysis showing competitors most often fail on implicit constraints and logical relations that the framework targets directly.

Topic 2: An expert-knowledge-guided modelling engine. Topic 2 targets complex modelling tasks where key information is implicit and must be derived. Workforce scheduling is a representative case, because correct formulations depend on derived coverage structures, cyclic-horizon wrap-around logic, and additional components such as breaks and overtime.

We propose SMILO, an expert-knowledge-guided modelling engine that uses LLMs for language understanding, while keeping the mathematical structure under explicit control. The key idea is to encode reusable modelling expertise as modelling graphs and associated resources. Stage 1 identifies relevant modelling components. Stage 2 extracts instance-specific information via task-based prompts with dependencies and structured outputs. Stage 3 assembles the MILP deterministically using templates and rule-based derivations, which helps reduce hallucination by keeping mathematical construction under control.

SMILO is evaluated on two workforce scheduling problem types (daily shift scheduling and days-off scheduling) over 30 instances with five trials per instance, and it substantially outperforms one-step prompting and the strong automated baseline. The same architecture has also been instantiated for facility location as an additional problem type, achieving 100% accuracy on 15 instances. This highlights the applicability of the modelling-graph-guided design to other problem domains. It is potentially extensible to general mathematical programming.

Topic 3: Performance-aware automatic Benders decomposition configuration (ongoing). Even with correct models, large MILPs can remain computationally difficult. Benders decomposition can exploit separability, but its effectiveness depends on configuration decisions: which variables form the master problem, and

Table 1. Timeline for the PhD thesis topics

Project	Period	Status
Topic 1	Feb 2023 - Aug 2024	Under 2nd-round review
Topic 2	Mar 2024 - Sep 2025	Under 2nd-round review
Topic 3 and thesis writing	Aug 2025 - Aug 2026	Ongoing

how the residual structure is organised into subproblems. Prior automatic methods often infer block structure from graph representations or rank candidate decompositions using learning. A gap remains between structure and performance: partitions that look structurally plausible do not reliably yield the best computational performance.

Topic 3 targets learned, performance-aware Benders configuration for large-scale MILPs. The novelty is to reframe decomposition as a data-driven configuration problem where there is typically no unique correct decomposition scheme. Instead of relying on handcrafted structural heuristics, the proposed approach learns a performance-guided configuration generator. It proposes candidate decompositions guided by the instance structure and improves this generator using solver feedback. It aims to deliver instance-specific decompositions that improve computational progress.

3 Research plan and anticipated contributions

The remaining work focuses on finalising Topic 3 and consolidating the thesis, as outlined in Table 1. The anticipated thesis contributions are: (1) a reliable, controllable text-to-MILP synthesis framework for unstructured problems with logical relations; (2) an expert-knowledge-guided modelling engine based on modelling graphs that is applicable to complex domains with implicit requirements and derived quantities; and (3) a performance-aware Benders decomposition configuration approach that bridges the structure-performance gap.

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